

MA1521 L1 Week 02 Mon 20260817

Evaluate $\int \frac{x}{\sqrt{8-2x^2}} dx$.

Recall! $\frac{d}{dx} \sqrt{8-2x^2} = \frac{\frac{1}{2} \cdot (-4x)}{\sqrt{8-2x^2}} = \frac{-2x}{\sqrt{8-2x^2}}$

Hence $\int = \frac{-1}{2} \sqrt{8-2x^2} + \underline{\text{const.}}$
(using FTC)

Evaluate $\int \sec^8 x \tan x dx$.

Recall! $\frac{d}{dx} (\tan x = \frac{\sin x}{\cos x}) = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \underline{\sec^2 x} = \underline{1 + \tan^2 x}$

$$\frac{d}{dx} (\sec x = \frac{1}{\cos x}) = \frac{\sin x}{\cos^2 x} = \sec x \cdot \tan x$$

$$\frac{d}{dx} (\sec^n x = \frac{1}{\cos^n x}) = \frac{(-n) \cdot (-\sin x)}{\cos^{n+1} x} = n \sec^n x \tan x$$

Hence $\int = \frac{1}{8} \sec^8 x + \text{const}$

definite integral
 $\downarrow$
 Evaluate $\int_0^1 (3x+2)^n dx$ in terms of n . FTC

$$\frac{(3x+2)^{n+1}}{3(n+1)} \bigg|_{x=0}^{x=1} = \frac{1}{3(n+1)} (5^{n+1} - 2^{n+1})$$

$\nwarrow$
assumed pos. int.

Same way: Recall: $\frac{d}{dx} (\ln(x + \sqrt{x^2+1})) = \frac{1}{(x + \sqrt{x^2+1})} \cdot \left(1 + \frac{x}{\sqrt{x^2+1}}\right)$

so $\int_0^x \frac{dt}{\sqrt{t^2+1}} = \ln(x + \sqrt{x^2+1})$

$$= \frac{1}{(x + \sqrt{x^2+1})} \cdot \frac{\cancel{\sqrt{x^2+1}} + x}{\sqrt{x^2+1}} = \frac{1}{\sqrt{x^2+1}}$$

Evaluate $\int_0^x \frac{dt}{\sqrt{t^2+1}}$ as a function of $x > 0$.

Better way: Change of Integration Variables (Int. by Substitution)

Let $t = \tan \theta$ (equiv. $\theta = \arctan t$)

Then $\sqrt{t^2+1} = \sqrt{\sec^2 \theta} = \sec \theta$

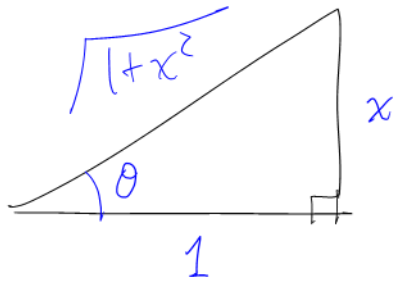
$$\begin{aligned} dt &= d(\tan \theta) = \frac{d \tan \theta}{d \theta} \cdot d\theta = (1 + \tan^2 \theta) d\theta \\ &= \sec^2 \theta d\theta \end{aligned}$$

$t = 0 \iff \theta = 0$

$t = x \iff \theta = \arctan x$

hence $\int_0^x \frac{dt}{\sqrt{t^2+1}} = \int_{\theta=0}^{\theta=\arctan x} \frac{1}{\sec \theta} \sec^2 \theta d\theta$

$= \int_0^{\arctan x} \sec \theta d\theta \stackrel{\text{review!}}{=} \ln(\sec \theta + \tan \theta) \Big|_{\theta=0}^{\theta=\arctan x}$



$$= \ln(\sqrt{1+x^2} + x) - 0$$

$$= \ln(\sqrt{1+x^2} + x) \quad \#$$

Gamma integral

Evaluate $\int_0^{\infty} x^3 e^{-x} dx$.

Give the answer as a **fraction** in lowest terms.

Beta integral

Evaluate $\int_0^1 x^2 (1-x)^5 dx$.

Give the answer as a **fraction** in lowest terms.

Evaluate $\int_5^7 \frac{1}{(x-4)(x-2)} dx$.

Give the **exact** answer using logarithms and fractions.

Start with $\int_0^{\infty} e^{-x} dx = [-e^{-x}]_0^{\infty} = 0 - (-1) = +1 = 0!$

$$\int_0^{\infty} x \cdot e^{-x} dx = \underbrace{[-x \cdot e^{-x}]_0^{\infty}}_{=0} - \int_0^{\infty} 1 \cdot (-e^{-x}) dx$$

$\int$ by parts

$$= + \int_0^{\infty} e^{-x} dx = +1 = 1!$$

$$\int_0^{\infty} x^2 \cdot e^{-x} dx = \underbrace{[-x^2 e^{-x}]_0^{\infty}}_0 - \int_0^{\infty} 2x^1 \cdot (-e^{-x}) dx$$

$\int$ by parts

$$= 2 \cdot \int_0^{\infty} x^1 \cdot e^{-x} dx = 2 = 2!$$

$$\int_0^{\infty} x^3 e^{-x} dx = \underbrace{[-x^3 e^{-x}]_0^{\infty}}_0 - \int_0^{\infty} 3x^2 \cdot (-e^{-x}) dx$$

$\int$ by parts

$$= + 3 \cdot \underbrace{\int_0^{\infty} x^2 e^{-x} dx}_2 = 6 = 3!$$

In general, $\int_0^{\infty} x^n e^{-x} dx = n!$ for any integer $n \geq 0$

$$\int_0^1 x^2 (1-x)^5 dx$$

$$\stackrel{q}{=} \int_{y=1}^{y=0} ((1-y)^2 \cdot y^5) (-dy)$$

Ch. of Var. $y = 1-x$ (so $x = 1-y$)

$$dy = \frac{d(1-x)}{dx} \cdot dx = -dx$$

When $x = \begin{cases} 1 \\ 0 \end{cases}$ then $y = \begin{cases} 0 \\ 1 \end{cases}$

$$= \int_{y=0}^{y=1} (1-y)^2 \cdot y^5 dy$$

$$= \int_0^1 (1-2y+y^2) \cdot y^5 dy$$

$$= \int_0^1 (y^5 - 2y^6 + y^7) dy = \left[\frac{y^6}{6} - 2 \cdot \frac{y^7}{7} + \frac{y^8}{8} \right]_0^1$$

$$= \frac{1}{6} - \frac{2}{7} + \frac{1}{8} = \frac{1}{168} = \frac{2! 5! 1!}{8!}$$

In general, if $a, b \geq 0$ are integers, then

$$B(a, b) = \int_0^1 x^a (1-x)^b dx = \frac{a! b! 1!}{(a+b+1)!} = \frac{1}{\binom{a+b+1}{a, b, 1}}$$

multinomial
coeff. $\swarrow$

$$\int_5^7 \frac{1}{(x-4)(x-2)} dx \stackrel{=}{=} \int_5^7 \left(\frac{\frac{1}{4-2}}{x-4} + \frac{\frac{1}{2-4}}{x-2} \right) dx$$

partial
fraction
expansion

$$= \frac{1}{2} \int_5^7 \left(\frac{1}{x-4} - \frac{1}{x-2} \right) dx$$

$$= \frac{1}{2} \ln \frac{x-4}{x-2} \Big|_5^7 = \frac{1}{2} \ln \frac{9}{5}$$

Supp. it takes 10^{-10} sec. to add each term of

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n} + \dots$$

Qn: How long to get answer > 1000 ?

Fact (exerc.): $\ln n < \underbrace{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n}} < 1 + \ln n$

For this sum to be > 1000

need $\ln n > 1000$ i.e. $n > e^{1000}$

Takes $10^{-10} \cdot e^{1000}$ sec. $\approx 1.97 \times 10^{424}$ sec.

$\Leftrightarrow 6.24 \times 10^{\underline{416}}$ yrs

Age of Univ. ≈ 13.8 billion yrs $\approx 1.38 \times 10^{\underline{10}}$ yrs

$$\int \frac{1}{1+x^2} dx \stackrel{?}{=} \arctan x$$

Ch. of Var

Evaluate $\int_1^{\sqrt{3}} \frac{dx}{x^2+1}$ $\int_{\pi/4}^{\pi/3} d\theta = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$

Ch. of Var. $x = \tan \theta$ ($\theta = \arctan x$)

$$dx = \left(\frac{d}{d\theta} \tan \theta \right) \cdot d\theta = (\sec^2 \theta) d\theta = (1 + \tan^2 \theta) d\theta = (1 + x^2) d\theta$$

∴ $\frac{dx}{x^2+1} = d\theta$

when $\begin{cases} x = \sqrt{3} \\ x = 1 \end{cases}$ have $\begin{cases} \theta = \frac{\pi}{3} \\ \theta = \frac{\pi}{4} \end{cases}$

If $F(x) = \int_x^{x^2} \tan t \, dt$, what is $F'(x)$?

FTC: If $F(x) = \int_0^x g(t) \, dt$

then $F'(x) = g(x)$

$= G(x^2) - G(x)$ where $G(x) := \int_0^x \tan t \, dt$

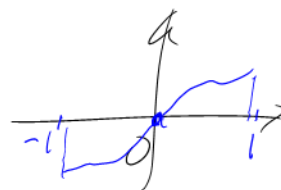
then $F'(x) = G'(x^2) \cdot 2x - G'(x)$

so FTC gives $G'(x) = \tan x$

$$= 2x \cdot \tan x^2 - \tan x$$

arc tan
↓

Evaluate $\int_{-1}^1 \frac{\tan^{-1} x}{x^2 + 1} dx$. $= 0$ bec. integrand $\frac{\tan^{-1} x}{x^2 + 1}$ is an odd function of x
 $\uparrow$ by symmetry of the integrand



partial frac.
↓

Evaluate $\int \frac{1}{(x-2)(x+3)} dx$. $= \int \left(\frac{\frac{1}{5}}{x-2} + \frac{-\frac{1}{5}}{x+3} \right) dx$

$= \frac{1}{5} \ln \left| \frac{x-2}{x+3} \right| + \text{const}$

partial frac.
↓

Evaluate $\int_1^2 \frac{x^2 + x + 1}{x(x+8)} dx$. $= \int_1^2 \left(1 + \frac{-7x+1}{x(x+8)} \right) dx = \int_1^2 \left(1 + \frac{\frac{1}{8}}{x} + \frac{-\frac{57}{8}}{x+8} \right) dx$
 $\uparrow$ long div of polynomials

$= 1 + \frac{1}{8} \ln 2 - \frac{57}{8} \ln \frac{10}{9}$

Evaluate $\int_0^{\pi/2} \frac{1}{(1 + \sin t)^2} dt$. later.

Evaluate $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$. $\xrightarrow{\text{l'Hôpital's rule}} \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0$

Evaluate $\int_e^{\infty} \frac{1}{x (\ln x)^2} dx = -(\ln x)^{-1} \Big|_{x=e}^{x=\infty} = -0 + \frac{1}{1} = 1$

Evaluate $\int \frac{e^{\tan x}}{\cos^2 x} dx$. $\text{Ch. of Var. } u = \tan x, \quad du = \frac{d(\tan x)}{dx} dx = \sec^2 x dx = \frac{1}{\cos^2 x} dx$
 $\Rightarrow \int e^u du = e^u + \text{const} = e^{\tan x} + \text{const}$

Give the most general answer as an **expression** in terms of x .

Evaluate $\int_0^{\pi/2} \frac{1}{(1 + \sin t)^2} dt$.

Heuristic: if integral is $\frac{\text{poly. in } \sin t, \cos t}{\text{poly. in } \sin t, \cos t}$

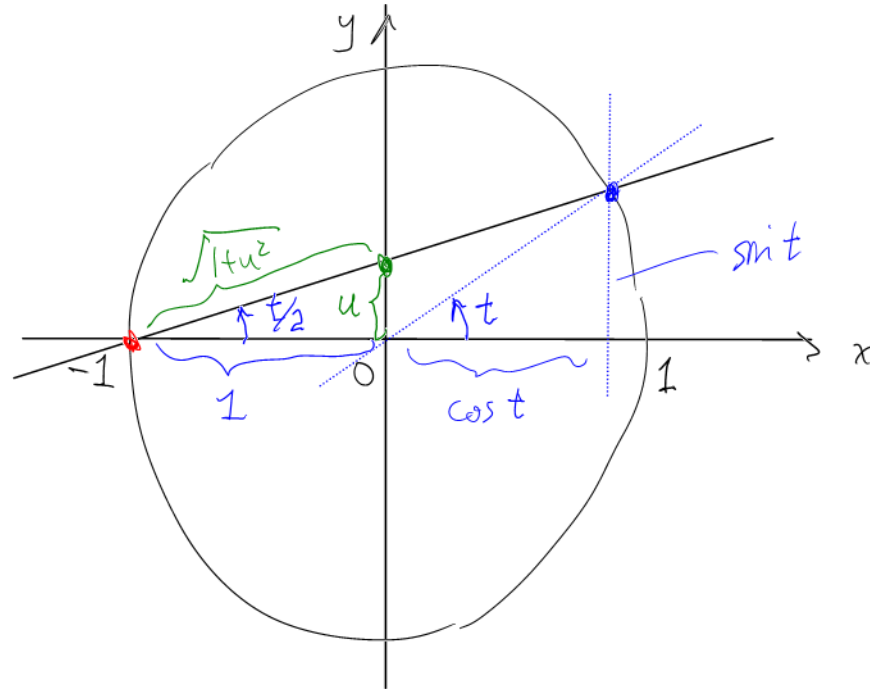
try Half-angle substitution $u := \tan \frac{t}{2}$

• $\sin t = \frac{2u}{1+u^2}$, $\cos t = \frac{1-u^2}{1+u^2}$

• $du = \frac{d}{dt} \left(\tan \frac{t}{2} \right) dt = \frac{1}{2} \sec^2 \frac{t}{2} \cdot dt = \frac{1+u^2}{2} dt$

so $dt = \frac{2}{1+u^2} du$

Half-angle substitution = stereographic projection of unit circle from $(-1, 0)$ to y-axis



• $\sin t = 2 \sin \frac{t}{2} \cdot \cos \frac{t}{2} = 2 \cdot \frac{u}{1+u^2} \cdot \frac{1}{1+u^2} = \frac{2u}{1+u^2}$

• $u := \tan \frac{t}{2} = \frac{\sin t}{1 + \cos t}$

so $1 + \cos t = \frac{2}{1+u^2}$, so $\cos t = \frac{2}{1+u^2} - 1 = \frac{1-u^2}{1+u^2}$

Evaluate $\int_0^{\pi/2} \frac{1}{(1 + \sin t)^2} dt.$ $\Rightarrow$ $\int_{u=0}^{u=1} \frac{1}{\left(1 + \frac{2u}{1+u^2}\right)^2} \frac{2}{1+u^2} du$

Ch. Var

$$u = \tan \frac{t}{2}$$

(above)

$$\Rightarrow 2 \cdot \int_{u=0}^{u=1} \frac{1+u^2}{(1+u)^4} du$$

Ch. Var

$$v = 1+u$$

$$dv = du$$

$$\rightarrow = 2 \cdot \int_{v=1}^{v=2} \frac{1+(v-1)^2}{v^4} dv = 2 \int_1^2 \frac{v^2 - 2v + 2}{v^4} dv$$

$$= 2 \int_1^2 \left(\frac{1}{v^2} - \frac{2}{v^3} + \frac{2}{v^4} \right) dv$$

$$= 2 \left[-\frac{1}{v} + \frac{1}{v^2} - \frac{2}{3v^3} \right]_{v=1}^{v=2}$$

$$= 2 \left[\left(-\frac{1}{2} \right) - \left(-\frac{2}{3} \right) \right] = \frac{2}{3}$$